

Type 2EL/2DL/2LL/2KL Wi-FiTM + Bluetooth[®] + 802.15.4(2EL/2LL) Module

Unified Design Guide - Rev. 2.0

- NXP Chipset: IW612 / IW611 / IW610G / IW610F
- Design Name: Type 2EL/Type 2DL/Type 2LL/Type 2KL
- P/N: LBES5PL2EL-923 / LBEE5PL2DL-921/
LBES0ZZ2LL-001 / LBEE0ZZ2KL-001

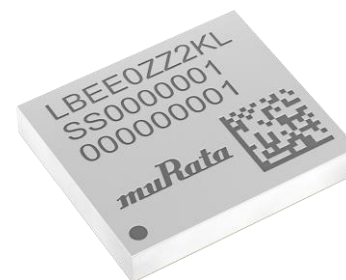
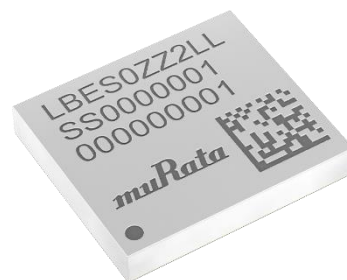


Table of Contents

1 Scope	4
2 Module Introduction	4
2.1 Features.....	4
2.1.1 Type 2EL / 2DL	4
2.1.2 Type 2LL / 2KL	4
2.2 Hardware Block Diagrams.....	5
3 Differences	9
3.1 Feature Differences.....	9
3.2 Pin Assignments	9
4 Reference Design	12
4.1 Wi-Fi, Bluetooth and 802.15.4	12
4.2 Wi-Fi and Bluetooth.....	13
4.3 Differences.....	14
4.3.1 PCM Interface, CONFIG_HOST	14
4.3.2 Software Reset Signal	14
4.3.3 Bluetooth Antenna.....	15
5 References	16
6 Technical Support Contacts.....	16
Revision History.....	17

Figures

Figure 1: Block Diagram - Type 2EL for Shared WLAN/Bluetooth/802.15.4 Antenna.....	5
Figure 2: Block Diagram - Type 2DL for Shared WLAN/Bluetooth Antenna	6
Figure 3: Block Diagram - Type 2LL for WLAN/Bluetooth/802.15.4 Antenna.....	7
Figure 4: Block Diagram - Type 2KL for WLAN/Bluetooth Antenna.....	8
Figure 5: Pin Assignments (Top View)	9
Figure 6: Reference schematics for Wi-Fi, Bluetooth and 802.15.4.....	12
Figure 7: Reference schematics for Wi-Fi, Bluetooth	13
Figure 8: PCM(2EL/2DL), CONFIG_HOST[2] (2EL/2KL)	14
Figure 9: Software Reset Signal for Bluetooth and 802.15.4	14
Figure 10: Bluetooth Antenna termination.....	15

Tables

Table 1: Document Conventions.....	3
------------------------------------	---

Table 2: Feature Differences.....	9
Table 3: Terminal Name Differences.....	10
Table 4: Reference Table	16
Table 5: List of Support Resources	16

About This Document

Type 2EL/2DL and Type2LL/2KL are pin to pin compatible modules. This Guide provides unified design guidance. Refer to [Datasheet](#)  for module specification.

Audience & Purpose

Intended audience includes any customer looking to integrate this module into their product. In particular, RF, hardware, systems, and software engineers.

Document Conventions

Table 1 describes the document conventions.

Table 1: Document Conventions

Conventions	Description
	Warning Note Indicates very important note. Users are strongly recommended to review.
	Info Note Intended for informational purposes. Users should review.
	Menu Reference Indicates menu navigation instructions. Example: Insert → Tables → Quick Tables → Save Selection to Gallery 
	External Hyperlink This symbol indicates a hyperlink to an external document or website. Example: Murata  Click on the text to open the external link.
	Internal Hyperlink This symbol indicates a hyperlink within the document. Example: Scope  Click on the text to open the link.
Console input/output or code snippet	Console I/O or Code Snippet This text Style denotes console input/output or a code snippet.
<pre># Console I/O comment // Code snippet comment</pre>	Console I/O or Code Snippet Comment This text Style denotes a console input/output or code snippet comment. • Console I/O comment (preceded by "#") is for informational purposes only and does not denote actual console input/output. • Code Snippet comment (preceded by "//") may exist in the original code.

1 Scope

This guide provides detailed information for designing Type 2EL/Type 2DL/Type 2LL/Type 2KL within a single schematic and layout. For module specifications, please refer to [Datasheet](#) . For individual design guides and detailed performance information, please consult [Hardware Application Note](#) .

2 Module Introduction

Type 2EL/2DL is a small and high-performance module based on NXP IW612/IW611 combo chipset which supports Wi-Fi 802.11a/b/g/n/ac/ax + Bluetooth 5.4 BR/EDR/LE + 802.15.4 (2EL only) up to 600 Mbps PHY data rate on Wi-Fi and 3 Mbps PHY data rate on Bluetooth.

Type 2LL/2KL is a small and high-performance module based on NXP IW610G/IW610F combo chipset which supports Wi-Fi 802.11a/b/g/n/ac/ax(HE20) + Bluetooth 5.4 LE + 802.15.4 (2LL only) up to 115 Mbps PHY data rate on Wi-Fi and 2 Mbps PHY data rate on Bluetooth Low Energy.

2.1 Features

2.1.1 Type 2EL / 2DL

- WLAN 802.11a/b/g/n/ac/ax 1x1 SISO + Bluetooth Classic and Low Energy (Version 5.4) + 802.15.4 combo SMD module with NXP IW612.
- WLAN 802.11a/b/g/n/ac/ax 1x1 SISO + Bluetooth Classic and Low Energy (Version 5.4) combo SMD module with NXP IW611.
- Small size LGA package with resin molding and metal shielding.
- Host interfaces: SDIO 3.0 for WLAN; HCI UART, PCM, and I2S for Bluetooth; SPI for 802.15.4.
- WLAN MAC address and BD address and 802.15.4 MAC address are stored in OTP



802.15.4 only for 2EL/IW612

2.1.2 Type 2LL / 2KL

- WLAN 802.11a/b/g/n/ac/ax 1x1 SISO + Bluetooth Low Energy (Version 5.4) + 802.15.4 combo SMD module with NXP IW610G.
- WLAN 802.11a/b/g/n/ac/ax 1x1 SISO + Bluetooth Low Energy (Version 5.4) combo SMD module with NXP IW610F.
- Small size LGA package with resin molding and metal shielding.
- Host interfaces: SDIO 3.0 and USB 2.0 for WLAN; HCI UART and USB 2.0 for Bluetooth; SPI for 802.15.4.
- WLAN MAC address and BD address and 802.15.4 MAC address are stored in OTP



802.15.4 only for 2LL/IW610G

2.2 Hardware Block Diagrams

This section shows the block diagram. **Figure 1** shows Type2EL-shared antenna block diagram and **Figure 2** shows Type2DL-shared antenna block diagram. **Figure 3** shows Type2LL block diagram and **Figure 4** shows Type 2KL block diagram.

Figure 1: Block Diagram - Type 2EL for Shared WLAN/Bluetooth/802.15.4 Antenna

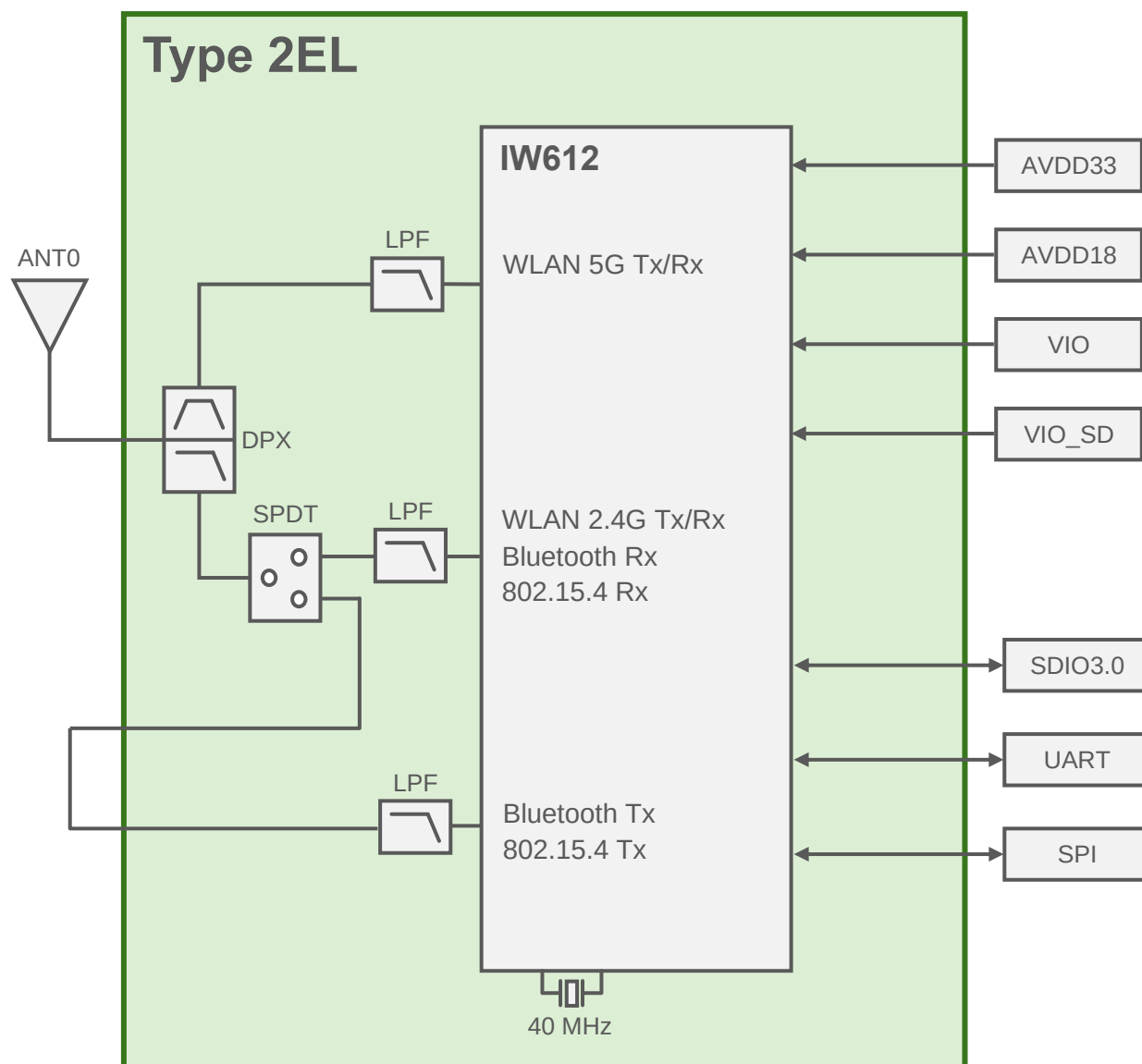


Figure 2: Block Diagram - Type 2DL for Shared WLAN/Bluetooth Antenna

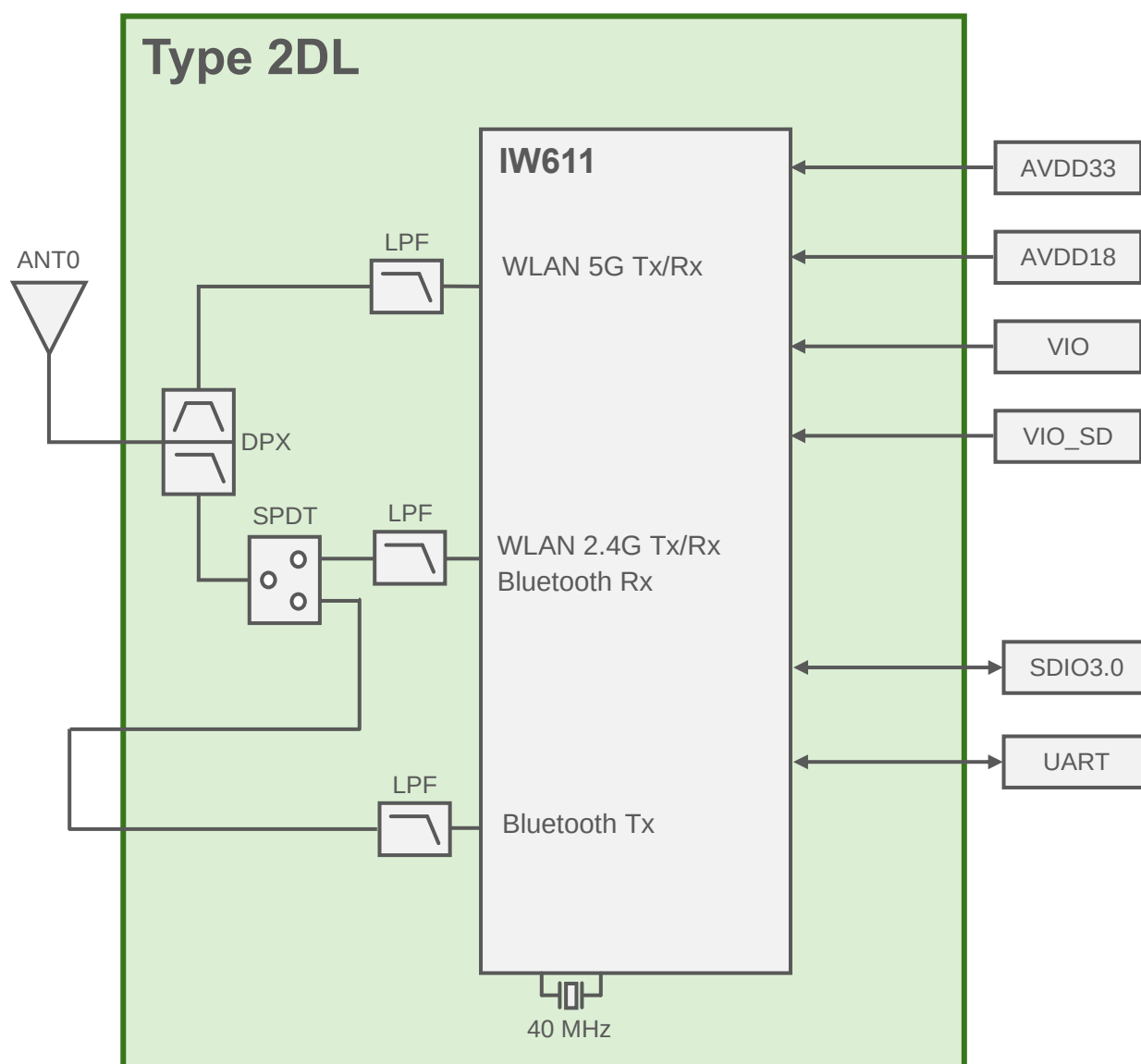


Figure 3: Block Diagram - Type 2LL for WLAN/Bluetooth/802.15.4 Antenna

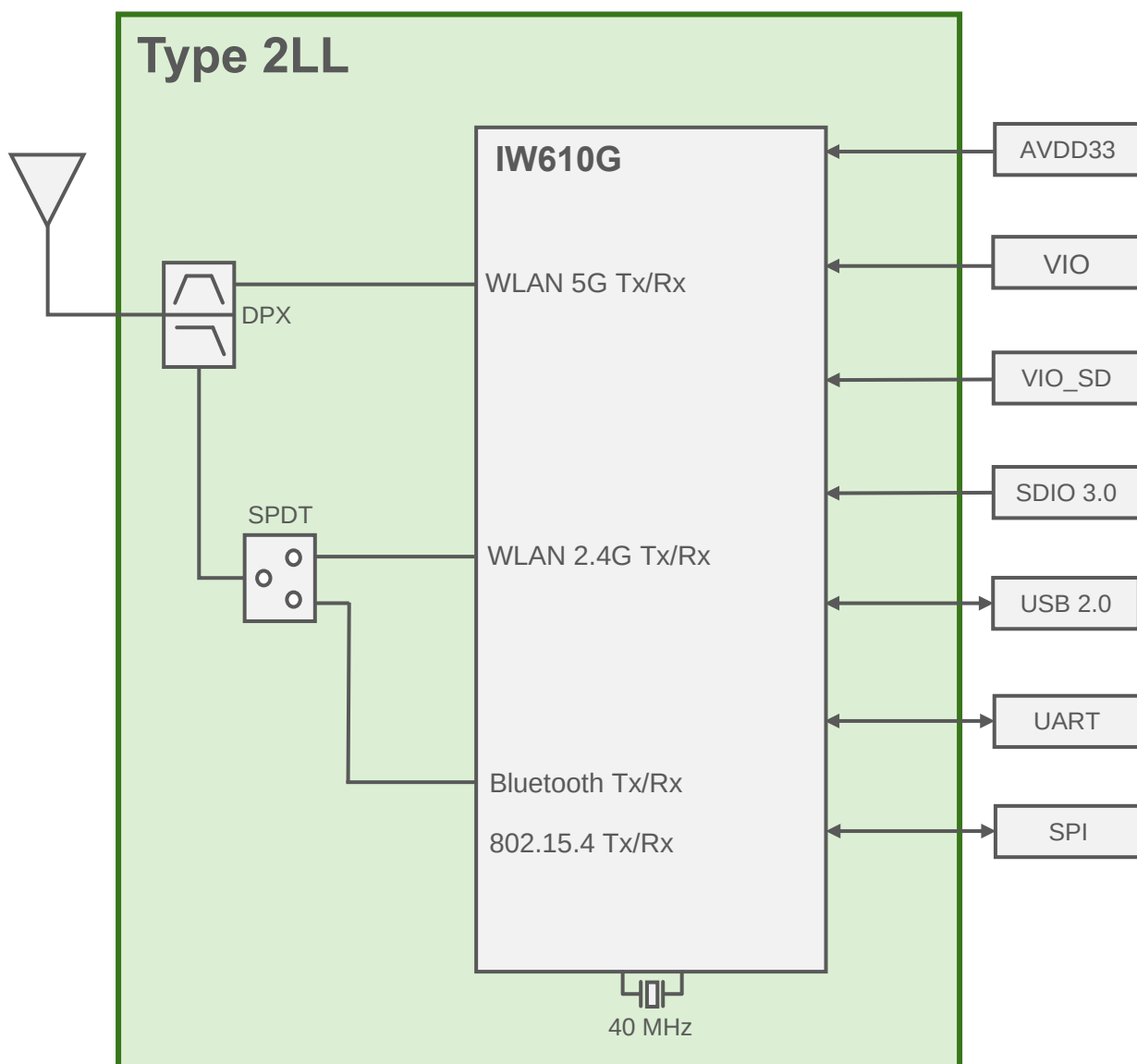
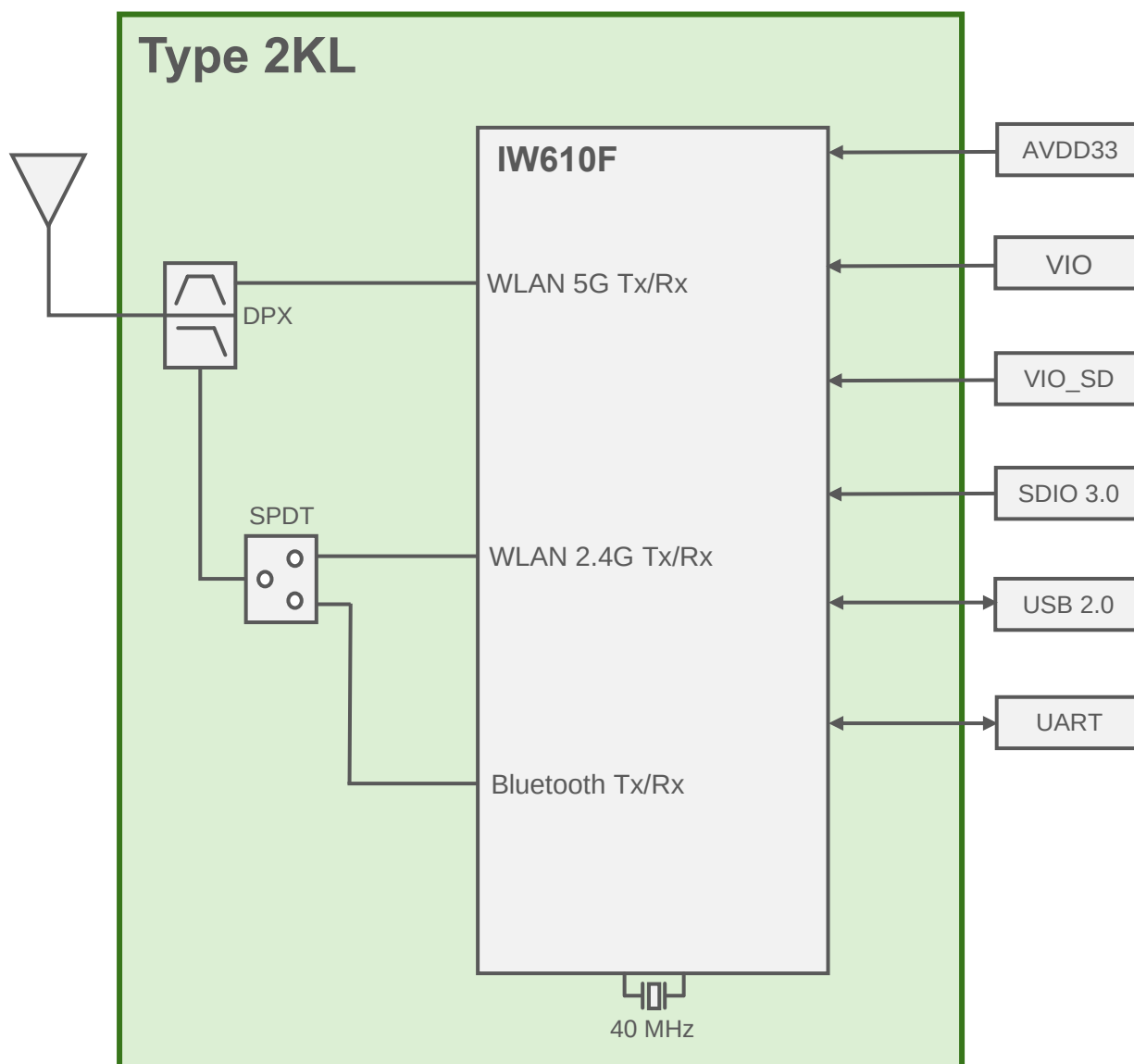


Figure 4: Block Diagram - Type 2KL for WLAN/Bluetooth Antenna



3 Differences

This section details differences for each module.

3.1 Feature Differences

Table 2 shows key differences for each module.

Table 2: Feature Differences

Features	Type 2EL (IW612)	Type 2DL (IW611)	Type 2LL (IW610G)	Type 2KL (IW610F)
40MHz(11n/ac/ax), 80MHz(11ac/ax) bandwidth operation	✓	✓	N/A	N/A
Bluetooth Classic operation	✓	✓	N/A	N/A
Bluetooth Low Energy operation	✓	✓	✓	✓
Bluetooth LE Audio	✓	✓	N/A	N/A
15.4 operation	✓	N/A	✓	N/A
Dedicated Bluetooth antenna operation	✓	✓	N/A	N/A
Dedicated Bluetooth/15.4 antenna operation	✓	N/A	N/A	N/A
USB interface for Wi-Fi and Bluetooth	N/A	N/A	✓	✓
PCM interface for Bluetooth Audio (Not LE Audio)	✓	✓	N/A	N/A

3.2 Pin Assignments

This section describes the pin assignments to terminals. **Figure 5** shows pin-assignment top view. All modules of Type 2EL, 2DL, 2LL and 2KL have the same pin-assignment. **Table 3** shows the terminal name.

Figure 5: Pin Assignments (Top View)

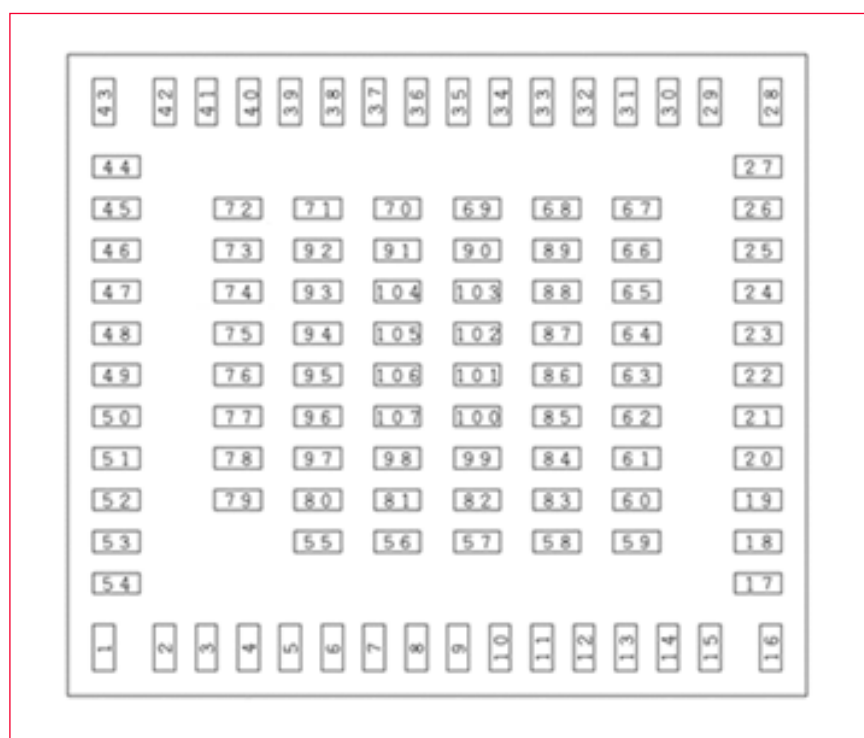


Table 3: Terminal Name Differences

No.	Type 2EL (IE612)	Type 2DL (IW611)	Type 2LL (IW610G)	Type 2KL (IW610F)
1	GND	GND	GND	GND
2	GND	GND	GND	GND
3	NC	NC	AVDD33_USB	AVDD33_USB
4	SPI_FRM	Reserved	SPI_FRMn	GPIO[8]
5	SPI_INT	Reserved	SPI_INT	CONFIG_HOST_BOOT[3]
6	SPI_RXD	Reserved	SPI_RXD	GPIO[7]
7	SPI_TXD	Reserved	SPI_TXD	GPIO[6]
8	SPI_CLK	Reserved	SPI_CLK	GPIO[9]
9	GND	GND	GND	GND
10	PDn	PDn	PDn	PDn
11	GND	GND	GND	GND
12	AVDD33_1	AVDD33_1	AVDD33	AVDD33
13	AVDD33_2	AVDD33_2	AVDD33	AVDD33
14	GND	GND	GND	GND
15	ANT0	ANT0	ANT0	ANT0
16	GND	GND	GND	GND
17	GND	GND	GND	GND
18	BT_15.4_IN	BT_IN	NC	NC
19	GND	GND	GND	GND
20	GND	GND	GND	GND
21	GND	GND	GND	GND
22	GND	GND	GND	GND
23	ANT1	ANT1	Reserved	Reserved
24	RF_CNTL4	RF_CNTL4	NC	NC
25	RF_CNTL3/ CONFIG_XOSC_SEL	RF_CNTL3/ CONFIG_XOSC_SEL	RF_CNTL3	RF_CNTL3
26	RF_CNTL1	RF_CNTL1	RF_CNTL2	RF_CNTL2
27	RF_CNTL0	RF_CNTL0	RF_CNTL0	RF_CNTL0
28	GND	GND	GND	GND
29	GND	GND	GND	GND
30	AVDD18_2	AVDD18_2	NC	NC
31	AVDD18_1	AVDD18_1	NC	NC
32	GND	GND	GND	GND
33	Reserved	Reserved	NC	NC
34	Reserved	Reserved	AVDD18_USB_OUT	AVDD18_USB_OUT
35	Reserved	Reserved	NC	NC
36	Reserved	Reserved	AVDD18_USB_IN	AVDD18_USB_IN
37	RST_IND	Reserved	NC	NC
38	IND_RST_15.4	Reserved	IND_RST_NB	IND_RST_NB
39	GND	GND	GND	GND
40	SD_VIO	SD_VIO	SD_VIO	SD_VIO
41	GND	GND	GND	GND
42	SD_CMD	SD_CMD	SD_CMD	SD_CMD
43	GND	GND	GND	GND
44	SD_CLK	SD_CLK	SD_CLK	SD_CLK
45	SD_DAT[1]	SD_DAT[1]	SD_DAT[1]	SD_DAT[1]
46	SD_DAT[3]	SD_DAT[3]	SD_DAT[3]	SD_DAT[3]
47	SD_DAT[2]	SD_DAT[2]	SD_DAT[2]	SD_DAT[2]
48	SD_DAT[0]	SD_DAT[0]	SD_DAT[0]	SD_DAT[0]
49	UART_TX	UART_TX	UART_SOUT	UART_SOUT
50	UART_CTS	UART_CTS	UART_CTSn	UART_CTSn

No.	Type 2EL (IE612)	Type 2DL (IW611)	Type 2LL (IW610G)	Type 2KL (IW610F)
51	UART_RX	UART_RX	UART_SIN	UART_SIN
52	UART_RTS	UART_RTS	UART_RTSn	UART_RTSn
53	GND	GND	GND	GND
54	VIO	VIO	VIO	VIO
55	CONFIG_HOST[0]	CONFIG_HOST[0]	CONFIG_HOST[0]	CONFIG_HOST[0]
56	CONFIG_HOST[1]	CONFIG_HOST[1]	CONFIG_HOST[1]	CONFIG_HOST[1]
57	PCM_CLK	PCM_CLK	NC	NC
58	PCM_MCLK	PCM_MCLK	Reserved	Reserved
59	PCM_DOUT	PCM_DOUT	Reserved	Reserved
60	PCM_DIN	PCM_DIN	CONFIG_HOST_BOOT[2]/ SD_VOLTAGE_SEL	CONFIG_HOST_BOOT[2]/ SD_VOLTAGE_SEL
61	PCM_SYNC	PCM_SYNC	NC	NC
62	GND	GND	GND	GND
63	IND_RST_WL	IND_RST_WL	IND_RST_WL	IND_RST_WL
64	IND_RST_BT	IND_RST_BT	IND_RST_BT	IND_RST_BT
65	Reserved	Reserved	EXT_GNT	EXT_GNT
66	Reserved	Reserved	EXT_REQ	EXT_REQ
67	GND	GND	GND	GND
68	GND	GND	GND	GND
69	WCI-2_SIN	WCI-2_SIN	WCI-2_SIN/ EXT_STATE	WCI-2_SIN/ EXT_STATE
70	WCI-2_SOUT	WCI-2_SOUT	WCI-2_SOUT/ EXT_FREQ	WCI-2_SOUT/ EXT_FREQ
71	GND	GND	GND	GND
72	SD_INT	SD_INT	SD_INT/USB_VBUS_ON	SD_INT/USB_VBUS_ON
73	WL_WAKE_OUT	WL_WAKE_OUT	WL_WAKE_OUT	WL_WAKE_OUT
74	WL_WAKE_IN	WL_WAKE_IN	WL_WAKE_IN	WL_WAKE_IN
75	BT15.4_WAKE_IN	BT_WAKE_IN	NB_WAKE_IN	NB_WAKE_IN
76	BT15.4_WAKE_OUT	BT_WAKE_OUT	NB_WAKE_OUT	NB_WAKE_OUT
77	NC	NC	USB_DP	USB_DP
78	NC	NC	USB_DM	USB_DM
79	NC	NC	Reserved	Reserved
80	NC	NC	EXT_PRI	EXT_PRI
81	GND	GND	GND	GND
- 107				

This section details reference schematics which the end user can leverage for designing their own hardware. For individual design guides and detailed performance information, please consult [Hardware Application Note](#) .

Figure 6 shows Wi-Fi / Bluetooth / 802.15.4 reference schematic.

- Standard: Wi-Fi, Bluetooth and 802.15.4
- Host Interface: SDIO for Wi-Fi, UART for Bluetooth, SPI for 802.15.4
- Antenna Configuration: Shared Bluetooth/802.15.4 Antenna Operation
- Audio interface: PCM for Bluetooth Audio (ex. A2DP)

(*) Must copy Murata's reference circuitry and layout.

(*) Strapping option of WLAN/BT interface

CONFIG_HOST [0] and [1] have an internal weak pull-up.

CONFIG_HOST [2] for Type 2LL has an internal weak pull-down.

CONFIG_HOST [0..1]	CONFIG_HOST[2]	WLAN	Bluetooth LE	802.15.4	Remarks
1 1	0	SDIO	UART	SPI	Default

(*) Please put damping resistor on SDIO signal lines. (SD_DAT0-3, SD_CMD, and SD_CLK)

Please adjust the resistor's value according to the design on the user's board.

(*) Type 2EL Shared Antenna: populate 10pF

Type 2LL : Depopulate 10pF

(*) Pin38 and 64 of Type 2LL connected inside module.

Type 2LL has only one reset signal for Bluetooth and 15.4

Type 2EL has individual reset signal for Bluetooth and 15.4.

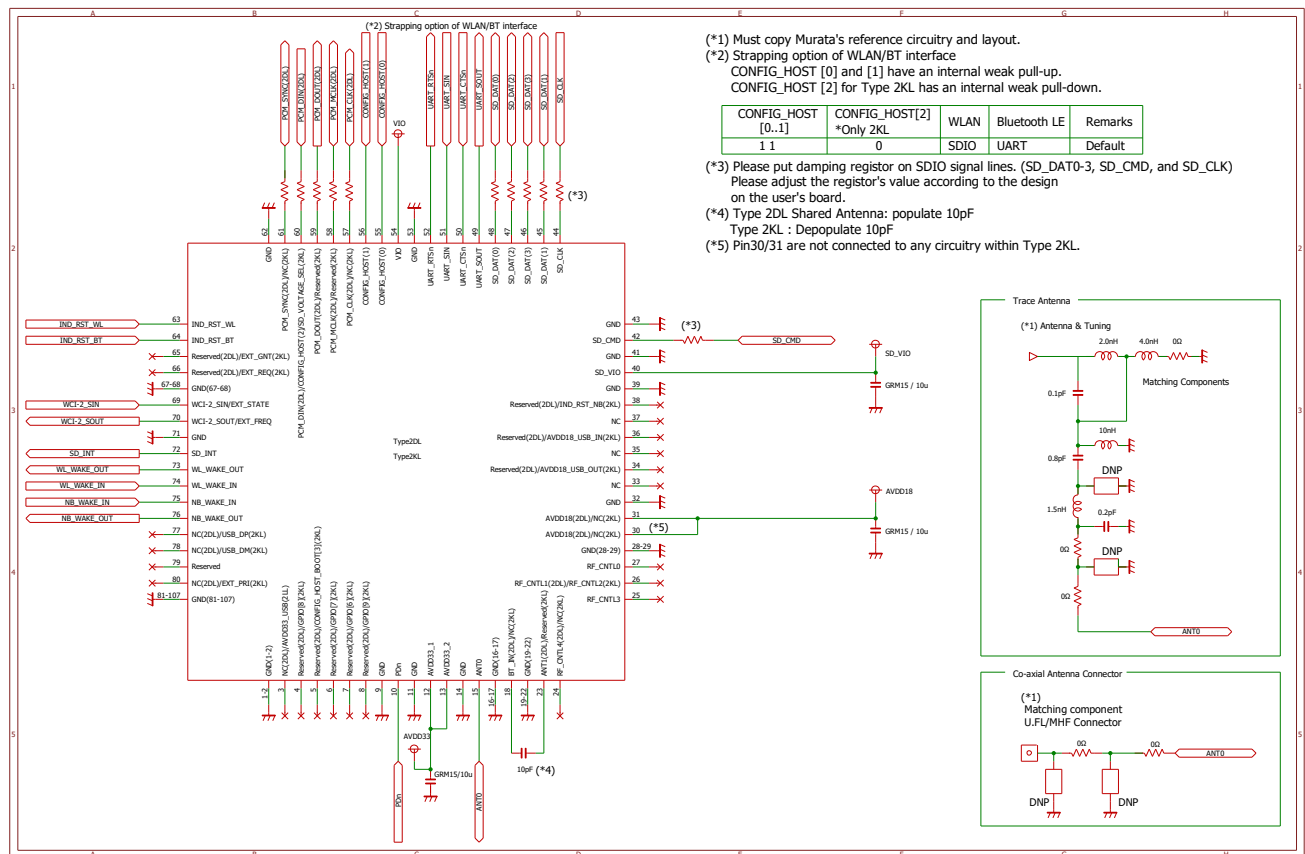
(*) Pin30/31 are not connected to any circuitry within Type 2LL.

4.2 Wi-Fi and Bluetooth

Figure 7 shows Wi-Fi / Bluetooth reference schematic.

- Wi-Fi and Bluetooth
- Host Interface: SDIO for Wi-Fi, UART for Bluetooth
- Antenna Configuration: Shared Bluetooth Antenna Operation
- Audio interface: PCM for Bluetooth Audio (ex. A2DP)

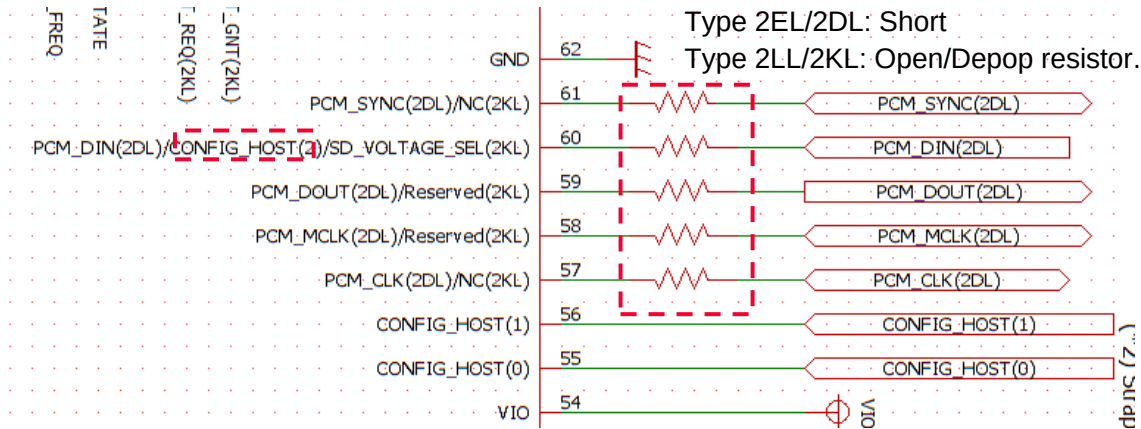
Figure 7: Reference schematics for Wi-Fi, Bluetooth



4.3 Differences

4.3.1 PCM Interface, CONFIG_HOST

Figure 8: PCM(2EL/2DL), CONFIG_HOST[2] (2EL/2KL)



CONFIG_HOST(2) of Type 2LL/2KL

- Host configuration option pin.
- IW610 has internal weak PD.
- PD(Low) is required for Type 2LL/2KL.

For 2KL/2LL cases, it is recommended to disconnect the Audio Interface PCM. If 2EL/2DL is to use PCM but is intended to share a common design with 2LL/2KL, please ensure that it can be set to a High-Z state in 2LL/2KL condition.

4.3.2 Software Reset Signal

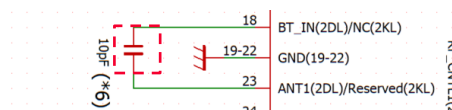
Figure 9: Software Reset Signal for Bluetooth and 802.15.4



Type 2EL has individual reset signal for Bluetooth and 15.4, but Type 2LL has only one reset signal for Bluetooth and 15.4. Type 2LL has IND_RST_NB (Reset for Bluetooth and 15.4) signal on the pin 38 and the pin 64, but these two pins are connected in the module. Bluetooth and 15.4 cannot be reset separately.

4.3.3 Bluetooth Antenna

Figure 10: Bluetooth Antenna termination



For Type 2KL and 2LL, the Bluetooth antenna inside the module is connected as a shared antenna with Wi-Fi.

For Type 2EL and 2DL, when using a shared antenna, it is necessary to connect Pin 18 and Pin 23 externally with a 10pF capacitor.

In the case of Type 2KL and 2LL, please remove the 10pF capacitor as a precaution.

5 References

Table 4 reviews all the key reference documents that the user may like to refer to.

Table 4: Reference Table

Support Site	Notes
Murata Type 2EL Module Datasheet 	Murata Type 2EL module datasheet (TYPE2EL.pdf)
Murata Type 2DL Module Datasheet 	Murata Type 2DL module datasheet (TYPE2DL.pdf)
Murata Type 2LL Module Datasheet 	Murata Type 2LL module datasheet (TYPE2LL.pdf)
Murata Type 2KL Module Datasheet 	Murata Type 2KL module datasheet (TYPE2KL.pdf)
Murata Type 2EL 2DL Hardware Application Note 	Murata Type 2EL 2DL hardware application note (type2el-2dl-hw-application-note.pdf)
Murata Type 2LL 2KL Hardware Application Note 	Murata Type 2LL 2KL hardware application note (type2ll-2kl-hw-application-note.pdf)



In case Murata website does not have the updated document, please refer to the [Connectivity Module](#)  page on the Murata Community Forum. This contains a pinned post with all the updated documents.

6 Technical Support Contacts

Table 5 lists all the support resources available for the Murata Wi-Fi/BT solution.

Table 5: List of Support Resources

Support Site	Notes
Murata Community Forum 	Primary support point for technical queries. This is an open forum for all customers. Registration is required.
Murata Module Landing Page 	No login credentials required. Murata documentation covering all Infineon-based Wi-Fi/BT modules is provided here.

Revision History

Revision	Date	Changed Item	Comment
1.0	May 27, 2025	First Issue	
2.0	Aug. 21, 2025	2.2 Hardware Block Diagrams	Removed PCM_SYNC from 2KL Block diagram
		3.1 Feature Differences	Status changed LE Audio/2KL from Applicable to N/A
		4.3.2 LE Audio Interface	Removed



Copyright © Murata Manufacturing Co., Ltd. All rights reserved. The information and content in this document are provided “as-is” with no warranties of any kind and are for informational purpose only. Data and information have been carefully checked and are believed to be accurate; however, no liability or responsibility for any errors, omissions, or inaccuracies is assumed.

The Bluetooth® word mark and logos are registered trademarks owned by Bluetooth SIG, Inc. Other brand and product names are trademarks or registered trademarks of their respective owners.

Specifications are subject to change without notice.